Top 41 CAT Selection With Condition Questions With Video Solutions

Questions

Instructions

Three reviewers Amal, Bimal, and Komal are tasked with selecting questions from a pool of 13 questions (Q01 to Q13). Questions can be created by external "subject matter experts" (SMEs) or by one of the three reviewers. Each of the reviewers either approves or disapproves a question that is shown to them. Their decisions lead to eventual acceptance or rejection of the question in the manner described below.

If a question is created by an SME, it is reviewed first by Amal, and then by Bimal. If both of them approve the question, then the question is accepted and is not reviewed by Komal. If both disapprove the question, it is rejected and is not reviewed by Komal. If one of them approves the question and the other disapproves it, then the question is reviewed by Komal. Then the question is accepted only if she approves it.

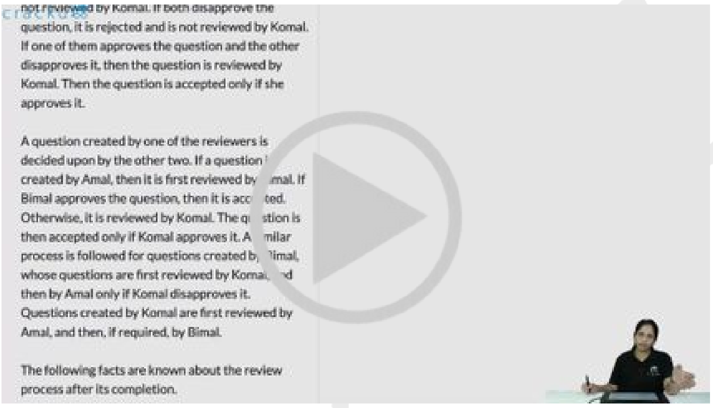
A question created by one of the reviewers is decided upon by the other two. If a question is created by Amal, then it is first reviewed by Bimal. If Bimal approves the question, then it is accepted. Otherwise, it is reviewed by Komal. The question is then accepted only if Komal approves it. A similar process is followed for questions created by Bimal, whose questions are first reviewed by Komal, and then by Amal only if Komal disapproves it. Questions created by Komal are first reviewed by Amal, and then, if required, by Bimal.

The following facts are known about the review process after its completion.

1. Q02, Q06, Q09, Q11, and Q12 were rejected and the other questions were accepted.
2. Amal reviewed only Q02, Q03, Q04, Q06, Q08, Q10, Q11, and Q13.
3. Bimal reviewed only Q02, Q04, Q06 through Q09, Q12, and Q13.
4. Komal reviewed only Q01 through Q05, Q07, Q08, Q09, Q11, and Q12.

Question 1

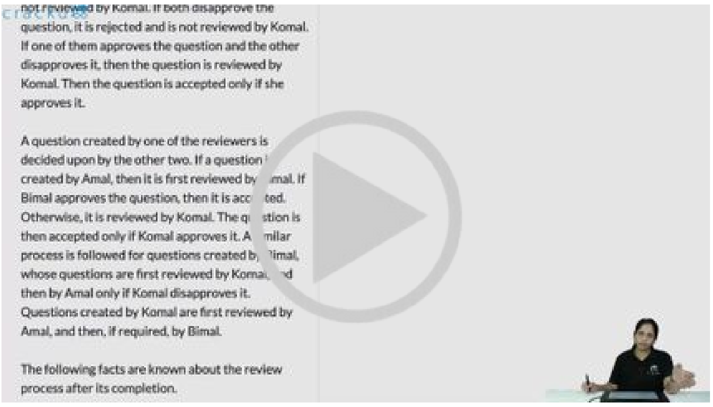
How many questions were DEFINITELY created by Amal?



VIDEO SOLUTION

Question 2

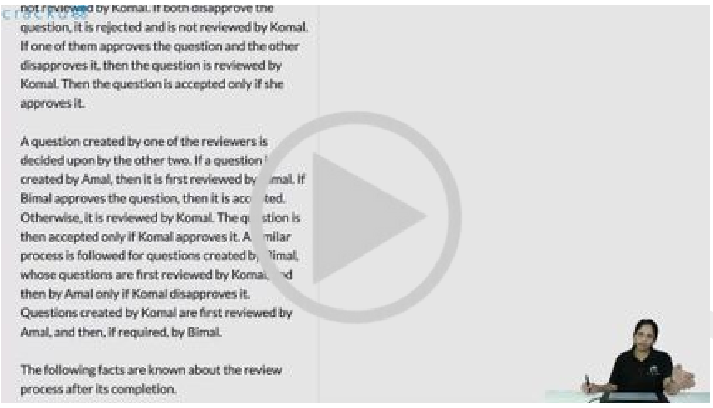
How many questions were DEFINITELY created by the SMEs?



VIDEO SOLUTION

Question 3

How many questions were DEFINITELY created by Komal?



VIDEO SOLUTION

Question 4

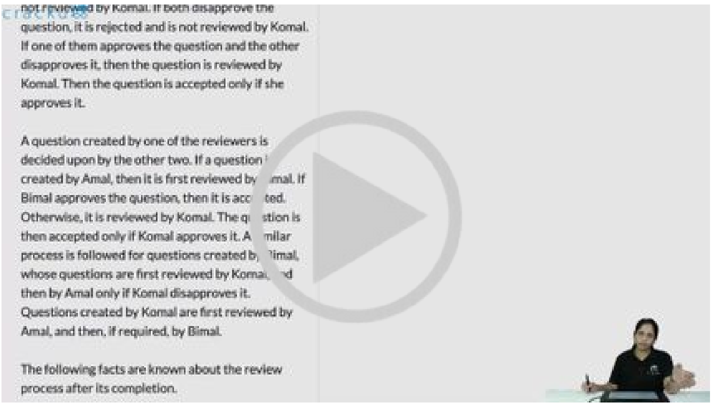
The approval ratio of a reviewer is the ratio of the number of questions (s)he approved to the number of questions (s)he reviewed. Which option best describes Amal's approval ratio?

A either 0.25 or 0.75

B 0.25

C lies between 0.25 and 0.50

D lies between 0.25 and 0.75



VIDEO SOLUTION

Question 5

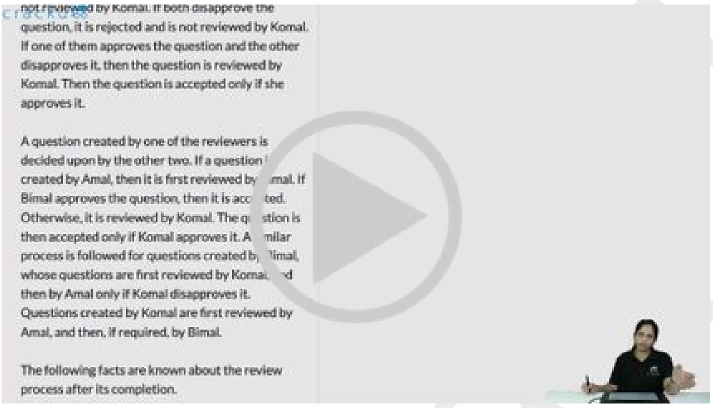
How many questions were DEFINITELY disapproved by Bimal?

A 3

B 4

C 7

D 5



VIDEO SOLUTION

Question 6

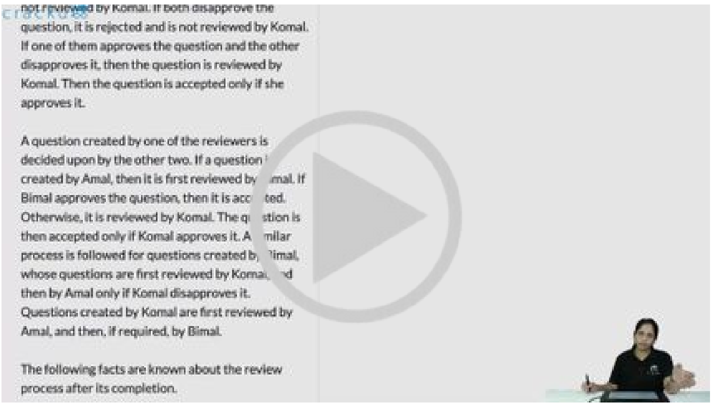
How many questions created by Amal or Bimal were disapproved by at least one of the other reviewers?

A 7

B 4

C 5

D 2



VIDEO SOLUTION

Instructions

Venkat, a stockbroker, invested a part of his money in the stock of four companies --- A, B, C and D. Each of these companies belonged to different industries, viz., Cement, Information Technology (IT), Auto, and Steel, in no particular order. At the time of investment, the price of each stock was Rs.100. Venkat purchased only one stock of each of these companies. He was expecting returns of 20%, 10%, 30%, and 40% from the stock of companies A, B, C and D, respectively. Returns are defined as the change in the value of the stock after one year, expressed as a percentage of the initial value. During the year, two of these companies announced extraordinarily good results. One of these two companies belonged to the Cement or the IT industry, while the other one belonged to either the Steel or the Auto industry. As a result, the returns on the stocks of these two companies were higher than the initially expected returns. For the company belonging to the Cement or the IT industry with extraordinarily good results, the returns were twice that of the initially expected returns. For the company belonging to the Steel or the Auto industry, the returns on announcement of extraordinarily good results were only one and a half times that of the initially expected returns. For the remaining two companies, which did not announce extraordinarily good results, the returns realized during the year were the same as initially expected.

Question 7

If Company C belonged to the Cement or the IT industry and did announce extraordinarily good results, then which of these statement(s) would necessarily be true?

I. Venkat earned not more than 36.25% return on average.

II. Venkat earned not less than 33.75% return on average.

III. If Venkat earned 33.75% return on average, Company A announced extraordinarily good results.

IV. If Venkat earned 33.75% return on average, Company B belonged either to Auto or to Steel Industry.

A I and II only

B II and IV only

C II and III only

D III and IV only

VIEW SOLUTION

Question 8

If Venkat earned a 38.75% return on average during the year, then which of these statement(s) would necessarily be true?

I. Company C belonged either to Auto or to Steel Industry.

II. Company D belonged either to Auto or to Steel Industry.

III. Company A announced extraordinarily good results.

IV. Company B did not announce extraordinarily good results.

A I and II only

B II and III only

C I and IV only

D II and IV only

VIEW SOLUTION

Question 9

What is the minimum average return Venkat would have earned during the year?

A 30%

B 31.25%

C 32.5%

D Cannot be determined

VIEW SOLUTION

Question 10

If Venkat earned a 35% return on average during the year, then which of these statements would necessarily be true?

I. Company A belonged either to Auto or to Steel Industry.

II. Company B did not announce extraordinarily good results.

III. Company A announced extraordinarily good results.

IV. Company D did not announce extraordinarily good results.

A I and II only

B II and III only

C III and IV only

D II and IV only

VIEW SOLUTION

Instructions

DIRECTIONS for the following three questions: Answer the questions on the basis of the information given below.

Colour	Rs / liter
RED	20
YELLOW	25
WHITE	15
ORANGE	22
PINK	18

Rang Barsey Paint Company (RBPC) is in the business of manufacturing paints. RBPC buys RED, YELLOW, WHITE, ORANGE, and PINK paints. ORANGE paint can be also produced by mixing RED and YELLOW paints in equal proportions. Similarly, PINK paint can also be produced by mixing equal amounts of RED and WHITE paints. Among other paints, RBPC sells CREAM paint, (formed by mixing WHITE and YELLOW in the ratio 70:30) AVOCADO paint (formed by mixing equal amounts of ORANGE and PINK paint) and WASHEDORANGE paint (formed by mixing equal amounts of ORANGE and WHITE paint). The following table provides the price at which RBPC buys paints.

Question 11

Assume that AVOCADO, CREAM and WASHEDORANGE each sells for the same price. Which of the three is the most profitable to manufacture?

A AVOCADO

B CREAM

C WASHEDORANGE

D Sufficient data is not available.

VIDEO SOLUTION

Question 12

The cheapest way to manufacture AVOCADO paint would cost

A Rs. 19.50 per litre.

B Rs. 19.75 per litre

C Rs. 20.00 per litre.

D Rs. 20.25 per litre.

VIDEO SOLUTION

Question 13

WASHEDORANGE can be manufactured by mixing

A CREAM and RED in the ratio 14:10.

B CREAM and RED in the ratio 3:1.

C YELLOW and PINK in the ratio 1:1.

D RED, YELLOW, and WHITE in the ratio 1:1:2.

VIDEO SOLUTION

Instructions

For the following questions answer them individually

Question 14

Four friends Ashok, Bashir, Chirag and Deepak are out for shopping. Ashok has less money than three times the amount that Bashir has. Chirag has more money than Bashir. Deepak has an amount equal to the difference of amounts with Bashir and Chirag. Ashok has three times the money with Deepak. They each have to buy at least one shirt, or one shawl, or one sweater, or one jacket that are priced Rs. 200, Rs. 400, Rs. 600, and Rs. 1,000 a piece respectively. Chirag borrows Rs. 300 from Ashok and buys a jacket. Bashir buys a sweater after borrowing Rs. 100 from Ashok and is left with no money. Ashok buys three shirts. What is the costliest item that Deepak could buy with his own money?

A A shirt

B A shawl

C A sweater

D A jacket

VIEW SOLUTION

Question 15

Eighty kilograms (kg) of store material is to be transported to a location 10 km away. Any number of couriers can be used to transport the material. The material can be packed in any number of units of 10,20 or 40kg. Couriers charges are Rs. 10 per hour. Couriers travel at the speed of 5 km/hr if carrying 10kg, at 2 km/hr if carrying 20kg and at 1 km/hr if carrying 40 kg. A courier cannot carry more than 40 kg of load. The minimum cost at which 80kg of store material can be transported to its distinction will be:

A Rs.180

B Rs.160

C Rs.140

D Rs.120

VIEW SOLUTION

Question 16

I brought 30 books on Mathematics, Physics, and Chemistry, priced at Rs.17, Rs.19, and Rs.23 per book respectively, for distribution among poor students of Standard X of a school. The physics books were more in number than the Mathematics books but less than the Chemistry books, the difference being more than one. The total cost amounted to Rs.620. How many books on Mathematics, Physics, and Chemistry could have been bought respectively?

A 5, 8, 17

B 5, 12, 13

C 5, 10, 15

D 5, 6, 19

VIEW SOLUTION

Question 17

A king has unflinching loyalty from eight of his ministers M1 to M8, but he has to select only four to make a cabinet committee. He decides to choose these four such that each selected person shares a liking with at least one of the other three selected. The selected persons must also hate at least one of the likings of any of the other three persons selected.
M1 likes fishing and smoking, but hates gambling.
M2 likes smoking and drinking, but hates fishing.
M3 likes gambling, but hates smoking,
M4 likes mountaineering, but hates drinking,
M5 likes drinking, but hates smoking and mountaineering.
M6 likes fishing, but hates smoking and mountaineering.
M7 likes gambling and mountaineering, but hates fishing.
M8 likes smoking and gambling, but hates mountaineering.
Who are the four people selected by the king?

A M1, M2, M5 and M6

B M3, M4, M5 and M6

C M4, M5, M6 and M8

D M1, M2, M4 and M7

VIEW SOLUTION

Question 18

My bag can carry no more than ten books, I must carry at least one book each of management, mathematics, physics and fiction. Also, for every management book I carry I must carry two or more fiction books, and for every mathematics book I carry I must carry two or more physics books. I earn 4, 3, 2 and 1 points for each management, mathematics, physics and fiction book, respectively, I carry in my bag. I want to maximise the points I can earn by carrying the most appropriate combination of books in my bag. The maximum points that I can earn are:

A 20

B 21

C 22

D 23

VIEW SOLUTION

Question 19

There are ten animals — two each of lions, panthers, bison, bears, and deer — in a zoo. The enclosures in the zoo are named X, Y, Z, P and Q and each enclosure is allotted to one of the following attendants: Jack, Mohan, Shalini, Suman and Rita. Two animals of different species are housed in each enclosure. A lion and a deer cannot be together. A panther cannot be with either a deer or a bison. Suman attends to animals from among bison, deer , bear and panther only. Mohan attends to a lion and a panther. Jack does not attend to deer, lion or bison. X, Y and Z are allotted to Mohan, Jack and Rita respectively. X and Q enclosures have one animal of the same species. Z and P have the same pair of animals. The animals attended by Shalini are:

A bear & bison

B bison & deer

C bear & lion

D bear & panther

VIEW SOLUTION

Question 20

I have a total of Rs. 1,000. Item A costs Rs. 110, item B costs Rs. 90, item C costs Rs. 70, item D costs Rs. 40 and item E costs Rs. 45. For every item D that I purchase, I must also buy two of item B. For every item A, I must buy one of item C. For every item E, I must also buy two of item D and one of item B. For every item purchased I earn 1,000 points and for every rupee not spent I earn a penalty of 1,500 points. My objective is to maximise the points I earn. What is the number of items that I must purchase to maximise my points?

A 13

B 14

C 15

D 16

VIEW SOLUTION

Instructions

K, L, M, N, P, Q, R, S, U and W are the only ten members in a department. There is a proposal to form a team from within the members of the department, subject to the following conditions:

1. A team must include exactly one among P,R and S.
2. A team must include either M or Q, but not both.
3. If a team includes K, then it must also include L, and vice versa.
4. If a team includes one among S, U and W, then it should also include the other two.
5. L and N cannot be members of the same team.
6. L and U cannot be members of the same team.

The size of a team is defined as the number of members in the team.

Question 21

What could be the size of a team that includes K?

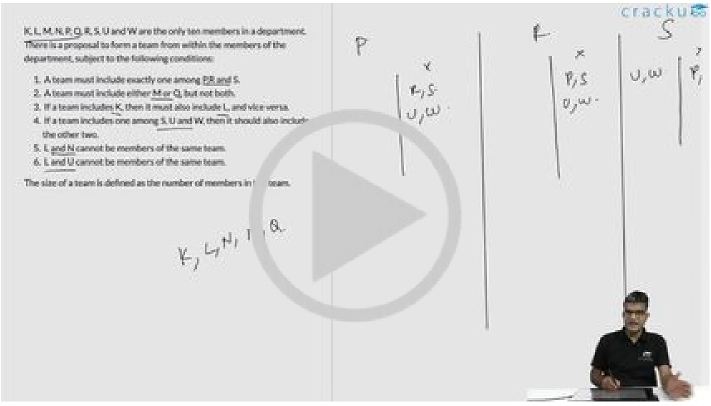
A 2 or 3

B 2 or 4

C 3 or 4

D Only 2

E Only 4



VIDEO SOLUTION

Question 22

Who cannot be a member of a team of size 3?

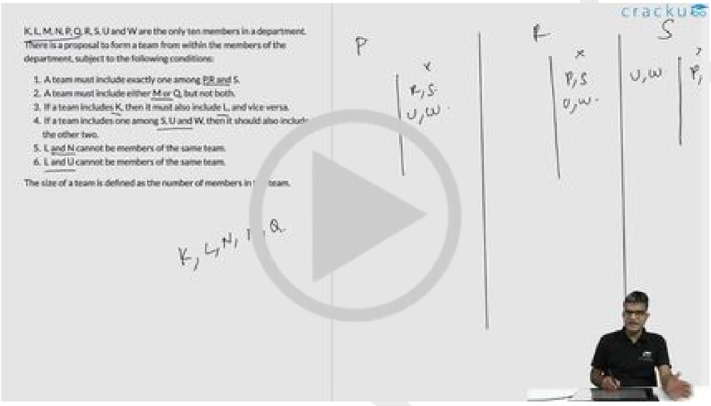
A L

B M

C N

D P

E Q



VIDEO SOLUTION

Question 23

Who can be a member of a team of size 5?

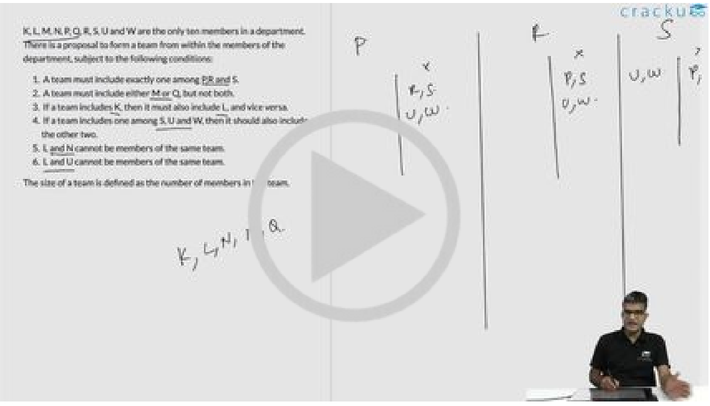
A K

B L

C M

D P

E R



VIDEO SOLUTION

Question 24

In how many ways a team can be constituted so that the team includes N?

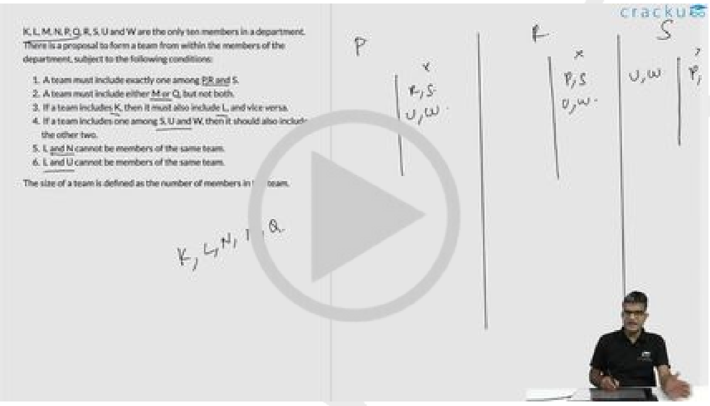
A 2

B 3

C 4

D 5

E 6



VIDEO SOLUTION

Question 25

What would be the size of the largest possible team?

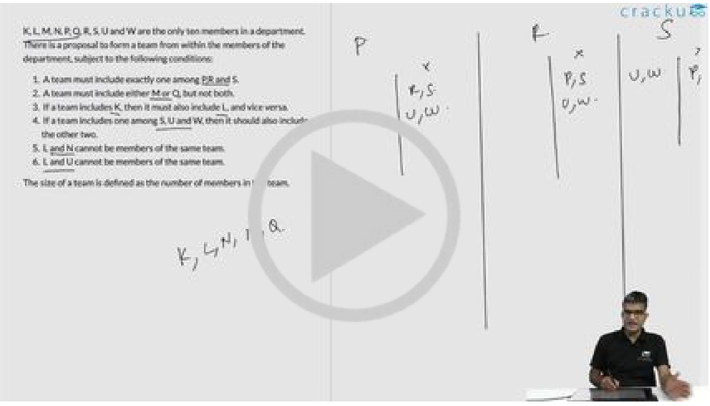
A 8

B 7

C 6

D 5

E cannot be determined



VIDEO SOLUTION

Instructions

Directions for the following three questions: Answer the questions based on the passage below.

A group of three or four has to be selected from seven persons. Among the seven are two women: Fiza and Kavita, and five men: Ram, Shyam, David, Peter and Rahim. Ram would not like to be in the group If Shyam is also selected. Shyam and Rahim want to be selected together in the group. Kavita would like to be in the group only if David is also there. David, if selected, would not like Peter in the group. Ram would like to be in the group only if Peter is also there. David insists that Fiza be selected in case he is there in the group.

Question 26

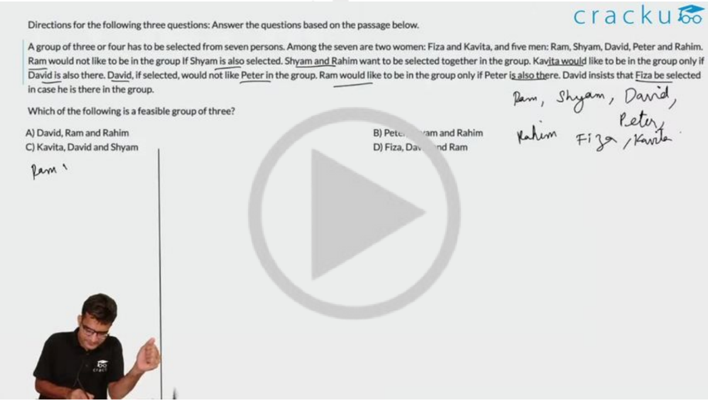
Which of the following is a feasible group in four?

A Ram, Peter, Fiza and Rahim

B Shyam, Rahim, Kavita and David

C Shyam, Rahim, Fiza and David

D Fiza, David, Ram and Peter



VIDEO SOLUTION

Question 27

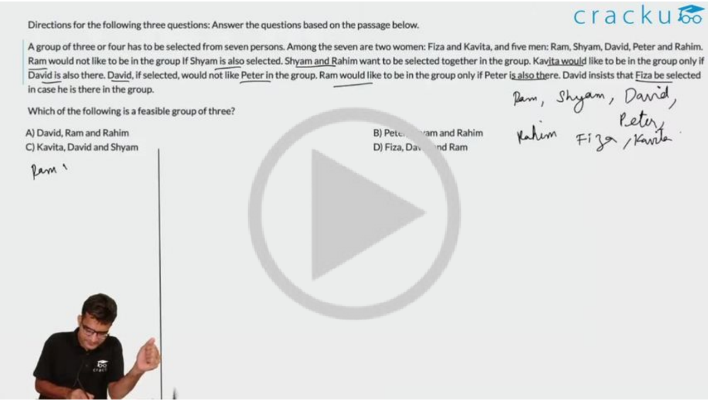
Which of the following is a feasible group of three?

A David, Ram and Rahim

B Peter, Shyam and Rahim

C Kavita, David and Shyam

D Fiza, David and Ram



VIDEO SOLUTION

Question 28

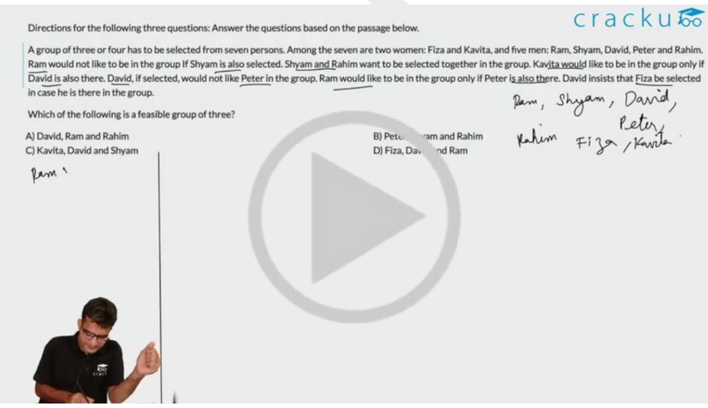
Which of the following statements is true?

A Kavita and Ram can be part of a group of four

B A group of four can have two women

C A group of four can have all four men

D None of these



Instructions

Two traders, Chetan and Michael, were involved in the buying and selling Of MCS shares over five trading days. At the beginning of the first day, the MCS share was priced at Rs 100, while at the end of the fifth day it was priced at Rs 110. At the end of each day, the MCS share price either went up by Rs 10, or else, it came down by Rs 10. Both Chetan and Michael took buying and selling decisions at the end of each trading day. The beginning price of MCS share on a given day was the same as the ending price of the previous day. Chetan and Michael started with the same number of shares and amount of cash, and had enough of both. Below are some additional facts about how Chetan and Michael traded over the five trading days.

• Each day if the price went up, Chetan sold 10 shares of MCS at the closing price. On the other hand, each day if the price went down, he bought 10 shares at the closing price.

• If on any day, the closing price was above Rs 110, then Michael sold 10 shares of MCS, while if it was below Rs 90, he bought 10 shares, all at the closing price.

Question 29

If Chetan ended up with Rs 1300 more cash than Michael at the end of day 5, what was the price of MCS share at the end of day 4?

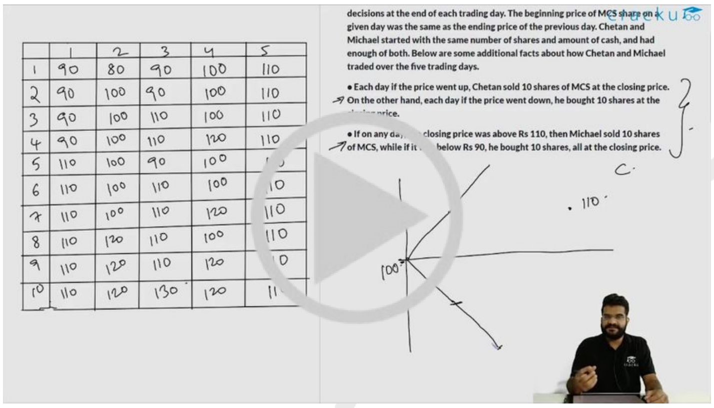
A Rs 90

B Rs 100

C Rs 110

D Rs 120

E Not uniquely determinable



Question 30

What could have been the maximum possible increase in combined cash balance of Chetan and Michael at the end of the fifth day?

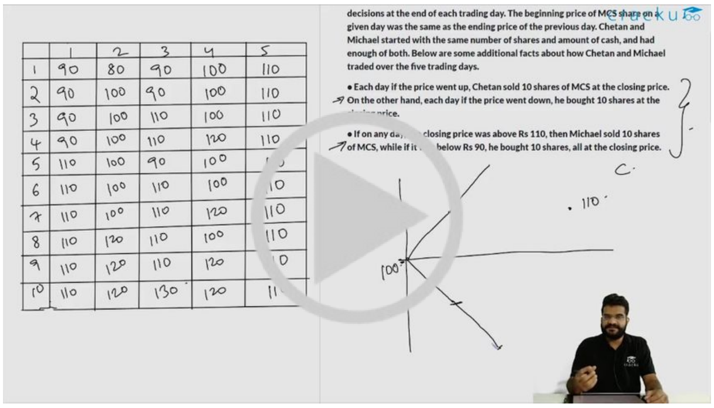
A Rs 3700

B Rs 4000

C Rs 4700

D Rs 5000

E Rs 6000



VIDEO SOLUTION

Question 31

If Chetan sold 10 shares of MCS on three consecutive days, while Michael sold 10 shares only once during the five days, what was the price of MCS at the end of day 3?

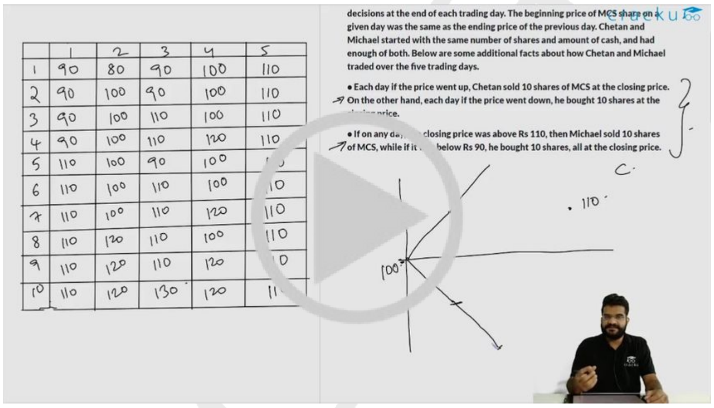
A Rs 90

B Rs 100

C Rs 110

D Rs 120

E Rs 130



VIDEO SOLUTION

Question 32

If Michael ended up with Rs 100 less cash than Chetan at the end of day 5, what was the difference in the number of shares possessed by Michael and Chetan (at the end of day 5)?

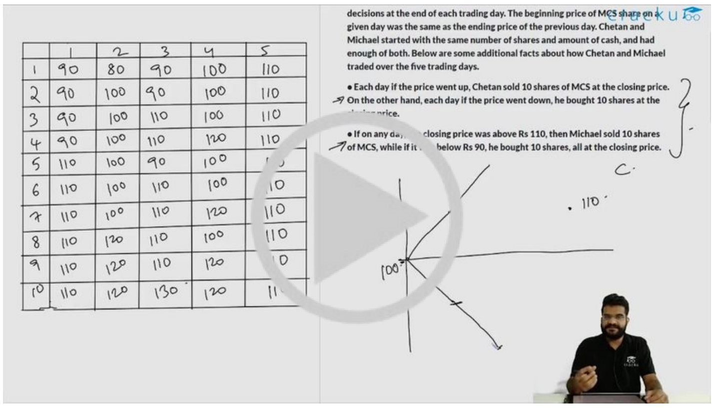
A Michael had 10 less shares than Chetan.

B Michael had 10 more shares than Chetan.

C Chetan had 10 more shares than Michael.

D Chetan had 20 more shares than Michael.

E Both had the same number of shares.



VIDEO SOLUTION

Question 33

If Michael ended up with 20 more shares than Chetan at the end of day 5, what was the price of the share at the end of day 3?

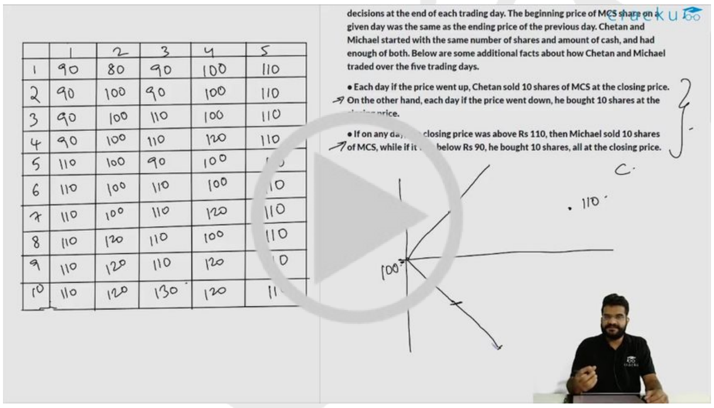
A Rs 90

B Rs 100

C Rs 110

D Rs 120

E Rs 130



VIDEO SOLUTION

Instructions

Directions for the following four questions: Answer the questions on the basis of the information given below.

Twenty one participants from four continents (Africa, Americas, Australasia, and Europe) attended a United Nations conference. Each participant was an expert in one of four fields, labour, health, population studies, and refugee relocation. The following five facts about the participants are given.

(a) The number of labour experts in the camp was exactly half the number of experts in each of the three other categories.

(b) Africa did not send any labour expert. Otherwise, every continent, including Africa, sent at least one expert for each category.

(c) None of the continents sent more than three experts in any category.

(d) If there had been one less Australasian expert, then the Americas would have had twice as many experts as each of the other continents.

(e) Mike and Alfanso are leading experts of population studies who attended the conference. They are from Australasia.

Question 34

If Ramos is the lone American expert in population studies, which of the following is NOT true about the numbers of experts in the conference from the four continents?

A There is one expert in health from Africa.

B There is one expert in refugee relocation from Africa.

C There are two experts in health from the Americas.

D There are three experts in refugee relocation from the Americas.

VIDEO SOLUTION

Question 35

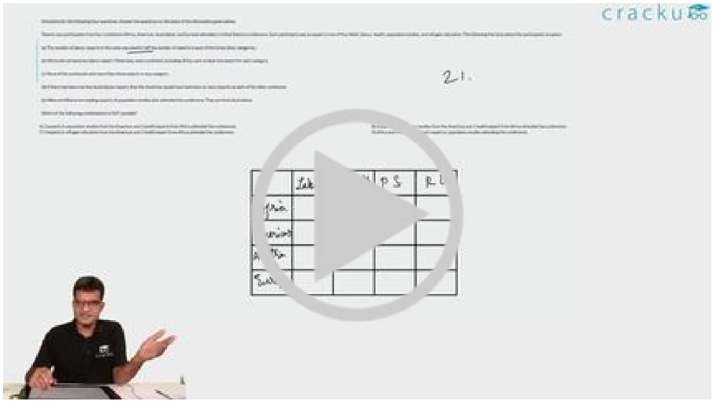
Which of the following combinations is NOT possible?

A 2 experts in population studies from the Americas and 2 health experts from Africa attended the conference.

B 2 experts in population studies from the Americas and 1 health expert from Africa attended the conference.

C 3 experts in refugee relocation from the Americas and 1 health expert from Africa attended the conference.

D Africa and America each had 1 expert in population studies attending the conference.



VIDEO SOLUTION

Question 36

Alex, an American expert in refugee relocation, was the first keynote speaker in the conference. What can be inferred about the number of American experts in refugee relocation in the conference, excluding Alex?

i. At least one

ii. At most two

A Only i and not ii

B Only ii and not I

C Both i and ii

D Neither i nor ii

VIDEO SOLUTION

Question 37

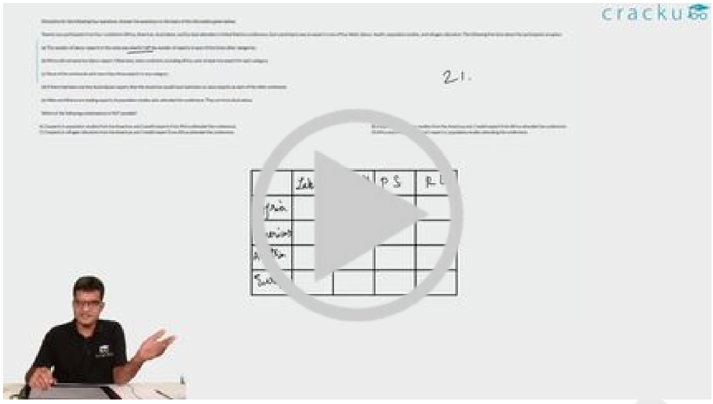
Which of the following numbers cannot be determined from the information given?

A Number of labour experts from the Americas.

B Number of health experts from Europe.

C Number of health experts from Australasia.

D Number of experts in refugee relocation from Africa.



VIDEO SOLUTION

Instructions

In the table below the check marks indicate all languages spoken by five people: Paula, Quentin, Robert, Sally and Terence. For example, Paula speaks only Chinese and English.

	Arabic	Basque	Chinese	Dutch	English	French
Paula			✓		✓	
Quentin				✓	✓	
Robert	✓					✓
Sally		✓			✓	
Terence			✓			✓

These five people form three teams, Team 1, Team 2 and Team 3. Each team has either 2 or 3 members. A team is said to speak a particular language if at least one of its members speak that language.
The following facts are known.
(1) Each team speaks exactly four languages and has the same number of members.
(2) English and Chinese are spoken by all three teams, Basque and French by exactly two teams and the other languages by exactly one team.
(3) None of the teams include both Quentin and Robert.
(4) Paula and Sally are together in exactly two teams.
(5) Robert is in Team 1 and Quentin is in Team 3.

Question 38

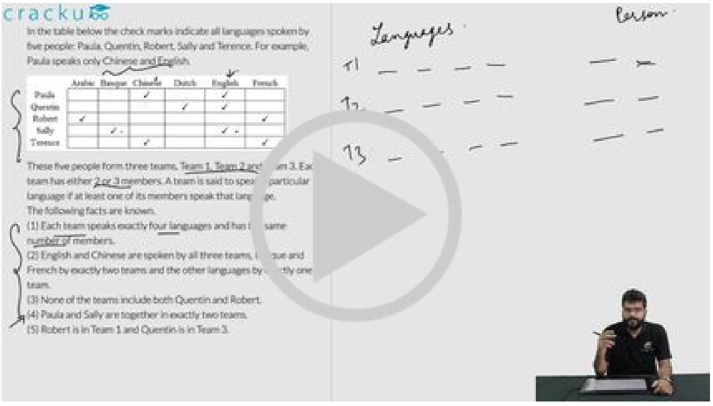
Who among the following four people is a part of exactly two teams?

A Paula

B Quentin

C Sally

D Robert



VIDEO SOLUTION

Question 39

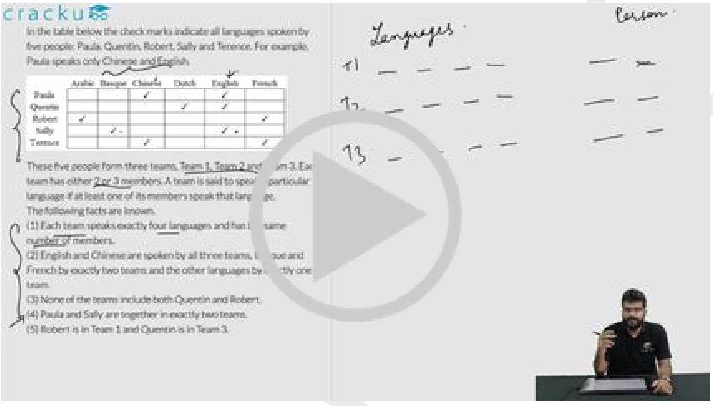
Who among the following four is not a member of Team 2?

A Paula

B Terence

C Quentin

D Sally



VIDEO SOLUTION

Question 40

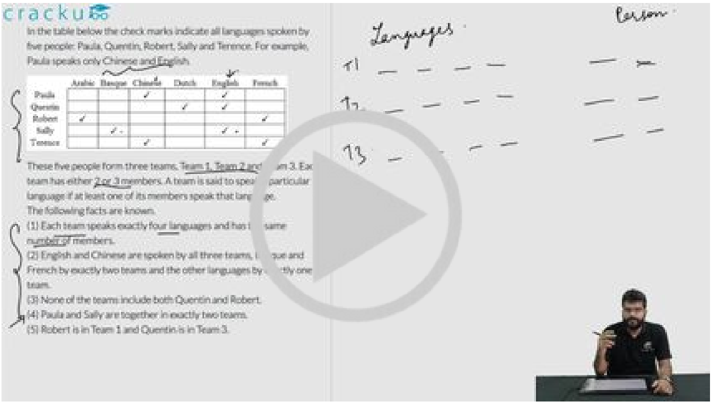
Who among the five people is a member of all teams?

A Terence

B Sally

C Paula

D No one



VIDEO SOLUTION

Question 41

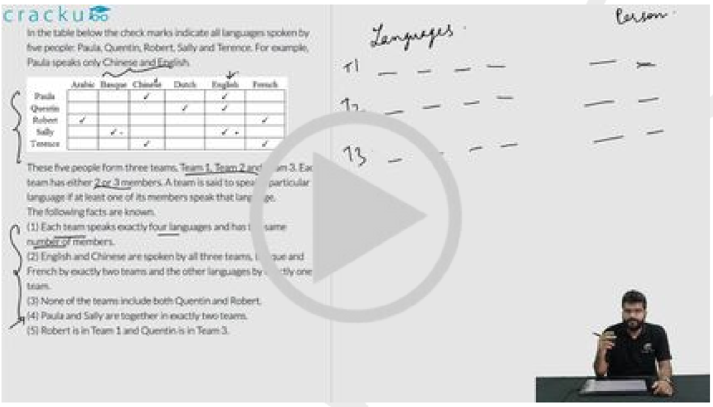
Apart from Chinese and English, which languages are spoken by Team 1?

A Arabic and French

B Basque and French

C Arabic and Basque

D Basque and Dutch



VIDEO SOLUTION

Answers

1.**3**	2.**3**	3.**1**	4.**D**	5.**B**	6.**C**	7.**B**	8.**C**
9.**A**	10.**B**	11.**B**	12.**B**	13.**D**	14.**B**	15.**B**	16.**C**
17.**D**	18.**C**	19.**C**	20.**B**	21.**E**	22.**A**	23.**C**	24.**E**
25.**D**	26.**C**	27.**B**	28.**D**	29.**B**	30.**D**	31.**C**	32.**E**
33.**A**	34.**C**	35.**D**	36.**C**	37.**D**	38.**C**	39.**C**	40.**C**
41.**A**							

Explanations

1.**3**

For a question created by externals(SME), the minimum number of reviews required is two and the maximum number of reviews is three to confirm the acceptance or the rejection.

For a question created by anyone among Amal, Bimal, Komal a question can be reviewed after one review or two reviews.

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Komal - Q01, Q02, Q03, Q04, Q05, Q07, Q08, Q09, Q11, and Q12.

Q01 is reviewed by only Komal and is accepted hence Bimal must have prepared the question as it was accepted in a single review. (Bimal)

Q02-is reviewed by Amal, Bimal, and Komal and was rejected this has a possibility of being prepared by an external person and faced rejection by one among Amal and Bimal and Komal. (SME).

Q03-Was accepted and reviewed by Amal, Komal and hence must have been prepared by Bimal(Bimal).

Q04- Accepted and reviewed by the three of them and hence must have been prepared by an external expert. (SME).

Q05-Only reviewed by Komal and is accepted and hence must be prepared by Bimal. (Bimal)

Q06-Rejected and reviewed by Amal and Bimal. Hence could have been prepared by Komal or External and rejected by Bimal and Amal. (Komal/SME).

Q07-Accepted and reviewed by Bimal and Komal. Hence must have been prepared by Amal. (Amal).

Q08- Reviewed by Amal, Bimal, and Komal and is accepted. Hence must have been prepared by SME. (SME)

Q09- Reviewed by Bimal and Komal and is rejected. It must have been prepared by Amal and was rejected by both of them. (Amal).

Q10- Was reviewed by Amal and was accepted. Must have been prepared by Komal. (Komal)

Q11- Reviewed by Amal and Komal and was rejected. Must have been prepared by Bimal and rejected by both Amal and Komal. (Bimal)

Q12- Reviewed by Bimal and Komal and was rejected. Must have been prepared by Amal and was rejected by both Bimal and Komal. (Amal).

Q13- Reviewed by Amal and Bimal and was accepted. The question could have been prepared by Komal or SME. (Komal/SME).

Amal definitely prepared questions Q07, Q09, Q12.

VIDEO SOLUTION

2. **3**
For a question created by externals(SME), the minimum number of reviews required is two and the maximum number of reviews is three to confirm the acceptance or the rejection.

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For the questions :

Amal - Q02, Q03, Q04, Q06, Q08, Q10, Q11, and Q13.
Bimal - Q02, Q04, Q06, Q07, Q08, Q09, Q12, and Q13.

Komal - Q01, Q02, Q03, Q04, Q05, Q07, Q08, Q09, Q11, and Q12.

Q01 is reviewed by only Komal and is accepted hence Bimal must have prepared the question as it was accepted in a single review. (Bimal)

Q02-is reviewed by Amal, Bimal, and Komal and was rejected this has a possibility of being prepared by an external person and faced rejection by one among Amal and Bimal and Komal. (SME).

Q03-Was accepted and reviewed by Amal, Komal and hence must have been prepared by Bimal(Bimal).

Q04- Accepted and reviewed by the three of them and hence must have been prepared by an external expert. (SME).

Q05-Only reviewed by Komal and is accepted and hence must be prepared by Bimal. (Bimal)

Q06-Rejected and reviewed by Amal and Bimal. Hence could have been prepared by Komal or External and rejected by Bimal and Amal. (Komal/SME).

Q07-Accepted and reviewed by Bimal and Komal. Hence must have been prepared by Amal. (Amal).

Q08- Reviewed by Amal, Bimal, and Komal and is accepted. Hence must have been prepared by SME. (SME)

Q09- Reviewed by Bimal and Komal and is rejected. It must have been prepared by Amal and was rejected by both of them. (Amal).

Q10- Was reviewed by Amal and was accepted. Must have been prepared by Komal. (Komal)

Q11- Reviewed by Amal and Komal and was rejected. Must have been prepared by Bimal and rejected by both Amal and Komal. (Bimal)

Q12- Reviewed by Bimal and Komal and was rejected. Must have been prepared by Amal and was rejected by both Bimal and Komal. (Amal).

Q13- Reviewed by Amal and Bimal and was accepted. The question could have been prepared by Komal or SME. (Komal/SME).

Q02, Q04, Q08 were prepared SME

VIDEO SOLUTION

3. **1**
For a question created by externals(SME), the minimum number of reviews required is two and the maximum number of reviews is three to confirm the acceptance or the rejection.

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For the questions :

Amal - Q02, Q03, Q04, Q06, Q08, Q10, Q11, and Q13.
Bimal - Q02, Q04, Q06, Q07, Q08, Q09, Q12, and Q13.

Komal - Q01, Q02, Q03, Q04, Q05, Q07, Q08, Q09, Q11, and Q12.

Q01 is reviewed by only Komal and is accepted hence Bimal must have prepared the question as it was accepted in a single review. (Bimal)

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Q04- Accepted and reviewed by the three of them and hence must have been prepared by an external expert. (SME).

Q05-Only reviewed by Komal and is accepted and hence must be prepared by Bimal. (Bimal)

Q06-Rejected and reviewed by Amal and Bimal. Hence could have been prepared by Komal or External and rejected by Bimal and Amal. (Komal/SME).

Q07-Accepted and reviewed by Bimal and Komal. Hence must have been prepared by Amal. (Amal).

Q08- Reviewed by Amal, Bimal, and Komal and is accepted. Hence must have been prepared by SME. (SME)

Q09- Reviewed by Bimal and Komal and is rejected. It must have been prepared by Amal and was rejected by both of them. (Amal).

Q10- Was reviewed by Amal and was accepted. Must have been prepared by Komal. (Komal)

Q11- Reviewed by Amal and Komal and was rejected. Must have been prepared by Bimal and rejected by both Amal and Komal. (Bimal)

Q12- Reviewed by Bimal and Komal and was rejected. Must have been prepared by Amal and was rejected by both Bimal and Komal. (Amal).

Q13- Reviewed by Amal and Bimal and was accepted. The question could have been prepared by Komal or SME. (Komal/SME).

Q10 was definitely prepared by Komal.

VIDEO SOLUTION

4. **D**

For a question created by externals(SME), the minimum number of reviews required is two and the maximum number of reviews is three to confirm the acceptance or the rejection.

For a question created by anyone among Amal, Bimal, Komal a question can be reviewed after one review or two reviews.

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For the questions :

Amal - Q02, Q03, Q04, Q06, Q08, Q10, Q11, and Q13.
Bimal - Q02, Q04, Q06, Q07, Q08, Q09, Q12, and Q13.

Komal - Q01, Q02, Q03, Q04, Q05, Q07, Q08, Q09, Q11, and Q12.

Q01 is reviewed by only Komal and is accepted hence Bimal must have prepared the question as it was accepted in a single review. (Bimal)

Q02-is reviewed by Amal, Bimal, and Komal and was rejected this has a possibility of being prepared by an external person and faced rejection by one among Amal and Bimal and Komal. (SME).

Q03-Was accepted and reviewed by Amal, Komal and hence must have been prepared by Bimal(Bimal).

Q04- Accepted and reviewed by the three of them and hence must have been prepared by an external expert. (SME).

Q05-Only reviewed by Komal and is accepted and hence must be prepared by Bimal. (Bimal)

Q06-Rejected and reviewed by Amal and Bimal. Hence could have been prepared by Komal or External and rejected by Bimal and Amal. (Komal/SME).

Q07-Accepted and reviewed by Bimal and Komal. Hence must have been prepared by Amal. (Amal).

Q08- Reviewed by Amal, Bimal, and Komal and is accepted. Hence must have been prepared by SME. (SME)

Q09- Reviewed by Bimal and Komal and is rejected. It must have been prepared by Amal and was rejected by both of them. (Amal).

Q10- Was reviewed by Amal and was accepted. Must have been prepared by Komal. (Komal)

Q11- Reviewed by Amal and Komal and was rejected. Must have been prepared by Bimal and rejected by both Amal and Komal. (Bimal)

Q12- Reviewed by Bimal and Komal and was rejected. Must have been prepared by Amal and was rejected by both Bimal and Komal. (Amal).

Q13- Reviewed by Amal and Bimal and was accepted. The question could have been prepared by Komal or SME. (Komal/SME).

Amal reviewed the questions : Q02, Q03, Q04, Q06, Q08, Q10, Q11, and Q13.

For Q2 - Amal can either approve or reject the question.

For Q3- Amal must approve the question.

For Q4 -Amal can either approve or reject the question.

For Q06 - Amal must reject the question.

For Q08 - Amal can either approve or reject the question.

For Q10 - Amal must approve the question.

For Q11 - Amal must reject the question.

For Q13- Amal can either approve or reject the question.

The approval ratio has a minimum of 2 questions only Q3, Q10, and a maximum of 6 questions Q02, Q3, Q04, Q08, Q10, Q13.

Hence 2/8 = 0.25 to 6/8 = 0.75

VIDEO SOLUTION

5.**B**

For a question created by externals(SME), the minimum number of reviews required is two and the maximum number of reviews is three to confirm the acceptance or the rejection.

For a question created by anyone among Amal, Bimal, Komal a question can be reviewed after one review or two reviews.

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Q12- Reviewed by Bimal and Komal and was rejected. Must have been prepared by Amal and was rejected by both Bimal and Komal. (Amal).

Q13- Reviewed by Amal and Bimal and was accepted. The question could have been prepared by Komal or SME. (Komal/SME).

Q06 irrespective of the question prepared by Komal/SME Bimal must reject the question.

Q07, Q09, Q12 are prepared by Amal and were reviewed by Bimal first and Komal next for this to happen Bimal must reject these questions.

Hence a total of four questions were rejected by Bimal.

VIDEO SOLUTION

6. **C**

For a question created by externals(SME), the minimum number of reviews required is two and the maximum number of reviews is three to confirm the acceptance or the rejection.

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Amal could have possibly created the questions Q07, Q09, Q12 of which all three of them were reviewed by two people each, and hence they must have been rejected by the first reviewer.

Bimal could have created the questions :

Q01, Q03, Q05, Q 11.

Q01, Q05 are reviewed only by Komal and hence did not have any rejection.

Q03, Q11 faced at least one rejection.

Hence a total of five questions.

VIDEO SOLUTION

7. **B**

Let us calculate the minimum average that Venkat can obtain in this case, and for this to happen, one and a half times the expected results must be obtained for company B. The value is calculated as follows,

	A	B	C	D	AVERAGE
Before results	20	10	30	40	25
After results	20	15	60	40	33.75

We can see that the minimum average obtained in the case is 33.75%, and Venkat cannot earn less than 33.75% in this case, making statement 2 true.

As we can see above, in the case of a 33.75% average return, Company A did not report extraordinary results, and Company C must belong to an auto or steel company, as it received one and a half times the expected results. So, statement 3 is false and 4 is true.

For the average to be maximum, one and a half must be obtained from company D.

Average maximum return = $\frac{20\ +\ 10\ +\ 60\ +\ 60}{4} = 37.5\%$

We can see that statement 1 is also false.

So, the correct statements are 2 and 4 only.

Hence, the correct answer is option B.

VIEW SOLUTION

8. **C**

For the average overall to be 38.75%, the sum of all the results has to be 38.75 * 4 = 155%. The sum is 155 in only a single case, when the results of D are doubled and the results of C become one and a half times the expected results. The calculations are as shown in the table below,

	A	B	C	D	Average
Before results	20	10	30	40	25
After results	20	10	45	80	38.75

To give a 38.75 % average return, we can see that B didn't give extraordinary returns, and we know that one of the auto or steel industries had one and a half times the expected results. As Company C got one and a half times the expected results, Company C has to be either the auto or steel industry.

So, the correct statements are 1 and 4 only.

Hence, the correct answer is option C.

VIEW SOLUTION

9. **A**

For the average profit to be minimum, the companies with returns higher than the initially expected returns have to be the companies with the least expected returns.

So, the company with twice the initially expected returns has to be the company that was expected to return 10%. So, the returns now become 20%.

The company with one and a half times the initially expected returns has to be the company with the next least expected results, which is 20%. So, the returns now become 30%.

The companies with 30% and 40% expected results got the same as the expected results.

The average overall profit can be calculated as,

$$\frac{20 + 30 + 30 + 40}{4} = 30\%$$

The minimum average return during the year for Venkat would be 30%.

Hence, the correct answer is option A.

VIEW SOLUTION

10. **B**

For the overall average to be 35%, there are two possible cases as shown below in the table,

Case 1: A gets double the expected results, and D gets one and a half times the expected results.

Case 2: C gets double the expected results, and A gets one and a half times the expected results.

		A	B	C	D	Average
Before Results		20	10	30	40	25
After Results	Case-1	40	10	30	60	35
	Case-2	30	10	60	40	35

In both cases, we can see that A must announce extraordinary results, and the results of B are as expected. But we can see that D has expected results in one case and had unexpected results in the other. Hence, statements 1 and 4 need not be necessarily true, and 2 and 3 must be true.

Hence, option B is the correct answer.

VIEW SOLUTION

11. **B**
Cream would be undoubtedly the most profitable as maximum amount of white paint is used in it and white is the cheapest out of all other paints. Hence option B.

VIDEO SOLUTION

12. **B**
AVOCADO paint can be manufactured by adding orange and pink in equal quantity. 0.5 ltr of orange would cost 11 and the cheapest way to make pink would be by mixing white and red , so the cost for 0.5 ltr of pink comes out to be 8.75 . SO total cost becomes 11+8.75 = 19.75 Rs. which is the cheapest.

VIDEO SOLUTION

13. **D**
WASHEDORANGE can be made by mixing orange and white , orange can be made by mixing equal quantities red and yellow which would be 1/2 of white quantity. Thus RED, YELLOW, and WHITE in the ratio 1:1:2 are needed.

VIDEO SOLUTION

14. **B**
According to given conditions Bashir started with 500Rs Chirag should have 700 + something , which when subtracted from 500 get money which Deepak has. Also Ashok should have < 1500 and his money should be 3 times that of deepak . Thus there;s only 1 possibility that Ashok has 1200rs and deepak has 400rs. So the costliest item that deepak can buy is shawl worth 400 rs.

VIEW SOLUTION

15. **B**
According to given conditions , for 10 kg it will take 2 hrs/10kg and total cost for 80 kg to be 160rs . for 20 kg it will take 5 hrs/20kg and total cost for 80 kg to be 200rs , for 40 kg it will take 10 hrs/40kg and total cost for 80 kg to be 200rs. So the cheapest is 160 rs.

VIEW SOLUTION

16. **C**
As it is mentioned that difference in number of books is more than one

Hence option B and D are cancelled.

Now total amount is = 620 = 17M + 19P + 23C

we can check by putting values from option A and C.

Option C satifies the above eq.,Hence our answer will be C

VIEW SOLUTION

17. **D**

	M1	M2	M3	M4	M5	M6	M7	M8
LIKE	F	S	G	M4	D	F	G	S
	S	D					M	G
DISLIKE	X G	X F	X S	X D	X S	XS		
					X M	XM	XF	XM

Looking at each option and the table. Only Option D satisfies the given condition that each selected person shares a liking with at least one of the other three selected. The selected persons must also hate at least one of the likings of any of the other three persons selected.

VIEW SOLUTION

18. **C**

The points would be maximum when the person carries 1 management, 2 fiction, 2 maths and 5 physics book.

Hence total points 4+2+6+10 = 22.

Hence option c.

VIEW SOLUTION

19. **C**

Correct arrange ment would be

NAME	JACK	Mohan	Shalini	Suman	Rita
ENCLOSURES	Y	X	Q	P	Z
ANIMALS	BEAR	LION	LION	BISON	BISON
	PANTHER	PANTHER	BEAR	DEER	DEER

Hence option C.

VIEW SOLUTION

20. **B**

According to given condition we find average costs for products bought.
If D is bought then - D +2*B . So we spend 40+180 =220 Rs for 3000 points. Hence, cost per 1000 points is 73.33
If A is bought then - A+C . So we spend 110 + 70 = 180 for 2000 points. Hence, cost per 1000 points is 90
If E is bought then - E+2*D+4*B+B . So we spend 45 + 80 + 360 + 90 = 575 for 8000 points. Hence, cost per 1000 points is 71.875.

If B is bought then I spend Rs 90 per 1000 points.

If C is bought then I spend Rs 70 per 1000 points.

To maximise points we need to select the item that costs the least per 1000 points. Hence, C costs the least per 1000 points. Hence, we should buy as many C items as possible.
Maximum C that can be bought is [1000/70] = 14 items and Rs 20 would be left over. However, the unspent

money would attract a penalty of 30000 points. Hence, instead of buying 14 C items, we should buy 13 C items and 1 B item so that total money spent is 1000.

VIEW SOLUTION

21. **E**
A team which has K should have L also.

Since L is there in the team, N and U should not be there in the team. Since U is not there in the team, S and W should not be there in the team.

So, the team will have K, L, one of P and R and one of M or Q.

So, the team size will be 4.

VIDEO SOLUTION

22. **A**
If L is in the team, the team should include K also. The team should have one among P, R and S and one among M and Q.

So, the team size cannot be 3 if L is in the team.

VIDEO SOLUTION

23. **C**
Out of P, R and S only 1 can be in the team. If S is there, U and W will also be there. So, P and R should not be in the team for its size to be maximum.

Out of M and Q, only 1 can be there.

If L is there in the team, N and U cannot be in the team.

If L is not there in the team, then K is also not there in the team but N and U can be in the team.

So, the maximum team size is 5 consisting of S, U, W, (M or Q), N.

So, M can be a member of team size 5.

VIDEO SOLUTION

24. **E**
Since N is in the team, L and K cannot be in the team.
The team can have one of M and Q. So, 2 ways of selection.
If the team has S, then it should have U and W as well.
If the team has no S, then it should have one of P or R.
So, the number of ways of forming the team is 2*(1+2) = 6 ways

VIDEO SOLUTION

25. **D**

Out of P, R and S only 1 can be in the team. If S is there, U and W will also be there. So, P and R should not be in the team for its size to be maximum.

Out of M and Q, only 1 can be there.

If L is there in the team, N and U cannot be in the team.

If L is not there in the team, then K is also not there in the team but N and U can be in the team.

So, the maximum team size is 5 consisting of S, U, W, (M or Q), N.

VIDEO SOLUTION

26. **C**

David and Peter cannot be selected together. So, option d) is not possible.
David and Fiza have to be selected together. So, option b) is ruled out.
Shyam and Rahim have to be selected together. So, option a) is also ruled out.
Option c) is the correct answer

VIDEO SOLUTION

27. **B**

Shyam and Rahim have to be selected together. This rules out options a) and c). Ram will be in the group only if Peter is also there. This rules out option d). Option b) is a feasible group

VIDEO SOLUTION

28. **D**

If Ram is selected, Peter also has to be selected. If Kavita is selected, David also has to be selected. But, Peter and David cannot be selected together. So, option a) is false.

If both the women are selected, David has to be selected. If David is selected, Peter cannot be selected. So, Ram also cannot be selected. Shyam and Rahim have to be selected together but there is room for only one more person. So, option b) cannot be true.

If David is selected in the group, Fiza has to be there. So, David cannot be in a group of all men. But, Ram and Shyam cannot be selected together. So, a group of all men is not possible. So option c is false.

VIDEO SOLUTION

29. **B**

case	day 1 end	day 2 end	day 3 end	day 4 end	day 5 end	chetan	michael
1	90	80	90	100	110	-10	10
2	90	100	110	120	110	-10	-10
3	90	100	90	100	110	-10	0
4	90	100	110	100	110	-10	0
5	110	100	90	100	110	-10	0
6	110	120	110	100	110	-10	-10
7	110	100	110	120	110	-10	-10
8	110	120	130	120	110	-10	-30
9	110	100	110	100	110	-10	0
10	110	120	110	120	110	-10	-20

The above table includes the values of the share price at the end of each day. Chetan and Michael column shows the number of shares at the end of 5th day with Chetan and Michael respectively. (-10 means he has sold 10 shares, +10 means he has bought 10 shares)

there are 10 different possible cases according to the initial and final share price.

Please note that Chetan will always be having -10 shares(10 shares sold) and 1300 as cash.

This is because Chetan buys for every fall in price and sells for every rise in price. But the fluctuation in share price is constant as it starts from 100 and closes at 110 on day 5. So, in total Chetan will always sell 10 shares in all 5 days combined.

As Chetan has sold 10 shares, he'll get 110*10 =1100 cash because of it in every case. Also, Chetan earns Rs. 200 in every case because of buying at low and selling at high. So total cash Chetan will always have after 5 days = 1100+200 =1300.

The question asks about the case where Michael ended up with Rs 1300 cash less than Chetan.

As Chetan will have exactly 1300 cash in every case; we have to look for the cases where Michael does not make any profit and also does not have any sold shares at the end of 5 days.

This is possible in 4 cases ie. 3,4,5 and 9.

In each of these cases, the price fluctuates in the range 90-110 which does not allow Michael to buy or sell any shares.

Also, in all these cases the price of MCS is 100 at the end of day 4.

So the answer is 100.

VIDEO SOLUTION

30. **D**

case	day 1 end	day 2 end	day 3 end	day 4 end	day 5 end	chetan	michael
1	90	80	90	100	110	-10	10
2	90	100	110	120	110	-10	-10
3	90	100	90	100	110	-10	0
4	90	100	110	100	110	-10	0
5	110	100	90	100	110	-10	0
6	110	120	110	100	110	-10	-10
7	110	100	110	120	110	-10	-10
8	110	120	130	120	110	-10	-30
9	110	100	110	100	110	-10	0
10	110	120	110	120	110	-10	-20

The above table includes the values of the share price at the end of each day. Chetan and Michael column shows the number of shares at the end of 5th day with Chetan and Michael respectively. (-10 means he has sold 10 shares, +10 means he has bought 10 shares)

there are 10 different possible cases according to the initial and final share price.

Please note that Chetan will always be having -10 shares(10 shares sold) and 1300 as cash.

This is because Chetan buys for every fall in price and sells for every rise in price. But the fluctuation in share price is constant as it starts from 100 and closes at 110 on day 5. So, in total Chetan will always sell 10 shares in all 5 days combined.

As Chetan has sold 10 shares, he'll get 110*10 =1100 cash because of it in every case. Also, Chetan earns Rs. 200 in every case because of buying at low and selling at high. So total cash Chetan will always have after 5 days = 1100+200 =1300.

The question asks about the case where there has been the maximum possible increase in the combined cash balance of Chetan and Michael at the end of the fifth day.

As we know chetan has 1300 cash in all the cases , so we have to maximize the case where Michael has the most cash.

Also, if we see clearly, the profit made by selling and buying is in hundreds while the cash received by selling the shares is much far in terms of cash.
As 10 shares sold give = 10*110 = 1100 cash. So we have to look at the case where Michael has sold most shares which is case 8.

In case 8 the price rises from 100 to 110 ->120 ->130 ->120 ->110

in this case 110(0 new shares)->120(Michael sold 10 shares)->130(Michael sold 10 shares)120(Michael sold 10 shares)->110(Michael does nothing) = 30 shares sold in total

Cash = 20 shares at 120 and 10 shares at 130 = 120*20+130*10 = 2400+1300 =3700

As Chetan has 1300 cash and Michael has 3700 cash;
total cash they have is : 3700+1300 =5000

VIDEO SOLUTION

31. **C**

case	day 1 end	day 2 end	day 3 end	day 4 end	day 5 end	chetan	michael
1	90	80	90	100	110	-10	10
2	90	100	110	120	110	-10	-10
3	90	100	90	100	110	-10	0
4	90	100	110	100	110	-10	0
5	110	100	90	100	110	-10	0
6	110	120	110	100	110	-10	-10
7	110	100	110	120	110	-10	-10
8	110	120	130	120	110	-10	-30
9	110	100	110	100	110	-10	0
10	110	120	110	120	110	-10	-20

The above table includes the values of the share price at the end of each day. Chetan and Michael column shows the number of shares at the end of 5th day with Chetan and Michael respectively. (-10 means he has sold 10 shares, +10 means he has bought 10 shares)

there are 10 different possible cases according to the initial and final share price.

The question asks about the case where Chetan has sold 3 times and Michael sells only once.

Starting with Michael, for exactly one sell, the price should touch 120 only once as Michael sells the share only at price greater than 110.

if the price touches 120 twice or more, Michael will sell the share more than once which is not a desirable case.

Also, chetan has to sell thrice consecutively which is only possible if the share price is 90 at one instance and rises to 120 in straight 3 days.

This is only possible in case 2. Hence the price on 3rd day's end in case 2 is 110.

VIDEO SOLUTION

32. **E**

case	day 1 end	day 2 end	day 3 end	day 4 end	day 5 end	chetan	michael
1	90	80	90	100	110	-10	10
2	90	100	110	120	110	-10	-10
3	90	100	90	100	110	-10	0
4	90	100	110	100	110	-10	0
5	110	100	90	100	110	-10	0
6	110	120	110	100	110	-10	-10
7	110	100	110	120	110	-10	-10
8	110	120	130	120	110	-10	-30
9	110	100	110	100	110	-10	0
10	110	120	110	120	110	-10	-20

The above table includes the values of the share price at the end of each day. Chetan and Michael column shows the number of shares at the end of 5th day with Chetan and Michael respectively. (-10 means he has sold 10 shares, +10 means he has bought 10 shares)

there are 10 different possible cases according to the initial and final share price.

Please note that Chetan will always be having -10 shares(10 shares sold) and 1300 as cash.

This is because Chetan buys for every fall in price and sells for every rise in price. But the fluctuation in share price is constant as it starts from 100 and closes at 110 on day 5. So, in total chetan will always sell 10 shares in all 5 days combined.

As chetan has sold 10 shares , he'll get 110*10 =1100 cash because of it in every case. Also chetan earns Rs. 200 in every case because of buying at low and selling at high. So total cash chetan will always have after 5 days = 1100+200 =1300.

The question asks about the case where Michael ended up with Rs 100 less cash than Chetan at the end of day 5.

So we have to look for the case where Michael has 1200 cash which is only possible when Michael has -10 number of shares in the end of day 5 and also has made profit of Rs. 100.

This is possible in case 7 as Michael has sold the shares at rs 120 but did not sell those shares as price never went below 90.

In the end Michael will have -10 shares at a price 110 which is Rs. 10 less than the price he sold at.

So he makes profit of 100 rs through it and will have the cash of 1100 through the sold shares.
Their difference in 7th case is 1300-1200=100

Also, in this case they have same number of shares sold at the end of the 5th day. So E is the answer.

VIDEO SOLUTION

33. **A**

case	day 1 end	day 2 end	day 3 end	day 4 end	day 5 end	chetan	michael
1	90	80	90	100	110	-10	10
2	90	100	110	120	110	-10	-10
3	90	100	90	100	110	-10	0
4	90	100	110	100	110	-10	0
5	110	100	90	100	110	-10	0
6	110	120	110	100	110	-10	-10
7	110	100	110	120	110	-10	-10
8	110	120	130	120	110	-10	-30
9	110	100	110	100	110	-10	0
10	110	120	110	120	110	-10	-20

The above table includes the values of the share price at the end of each day. Chetan and Michael column shows the number of shares at the end of 5th day with Chetan and Michael respectively. (-10 means he has sold 10 shares, +10 means he has bought 10 shares)

there are 10 different possible cases according to the initial and final share price.

Please note that Chetan will always be having -10 shares(10 shares sold) and 1300 as cash.

To have 20 more shares than Chetan, Michael has to buy 10 shares which is case 1.

In case 1, the share price at the end of day 3 is 90.

VIDEO SOLUTION

34. **C**
From the first condition a, assume the number of experts in labour = x, then the number of experts in each of the other three categories =2x
Now, x+2x+2x+2x+2x = 21 => x = 3 and 2x = 6
So the number of experts in labour = 3 and the number of experts in each of the three categories = 6
From condition d, if the number of experts of Astralaisa = y, then the number of experts of Americas = 2(y-1),

the number of experts of Europe = y-1, the number of experts of Africa= y-1
Now, y+y-1+y-1+2(y-1)=21 => 5y-4=21 => y=5
Hence the table can be filled as follows:

	Africa	America	Australasia	Europe	Total
Labour					3
Health					6
Population Studies					6
Refugee Location					6
Total	4	8	5	4	21

Using condition b, the number of labour experts of Africa = 0 and the number of labour experts of the rest of the continents will be 1 each.
Consider the column of Europe, since Europe sent at least 1 in each category, each category will have 1 expert. (Since the total sum is 4)
From the condition e, the population studies of Australasia will contain 2 experts. So all the other categories will have to take at 1 expert because the total sum is 5 and no value can be 0. The table can be filled as follows:

	Africa	America	Australasia	Europe	Total
Labour	0	1	1	1	3
Health			1	1	6
Population Studies			2	1	6
Refugee Location			1	1	6
Total	4	8	5	4	21

Now, according to the condition c, the possible solutions are as given in the tables below:

	Africa	America	Australasia	Europe	(Total)
Labour	0	1	1	1	3
health	2	2	1	1	6
Population Studies	1	2	2	1	6
Refugee Location	1	3	1	1	6
(Total)	4	8	5	4	21

	Africa	America	Australasia	Europe	(Total)
Labour	0	1	1	1	3
health	1	3	1	1	6
Population Studies	1	2	2	1	6
Refugee Location	2	2	1	1	6
(Total)	4	8	5	4	21

	Africa	America	Australasia	Europe	(Total)
Labour	0	1	1	1	3
Health	1	3	1	1	6
Population Studies	2	1	2	1	6
Refugee Location	1	3	1	1	6
(Total)	4	8	5	4	21

From the tables we can see that when there is only 1 population expert from America, there are 3 American Health Experts. Hence option C is not true.

VIDEO SOLUTION

35. **D**
From the first condition a, assume the number of experts in labour = x, then the number of experts in each of the other three categories =2x
Now, x+2x+2x+2x = 21 => x = 3 and 2x = 6
So the number of experts in labour = 3 and the number of experts in each of the three categories = 6

From condition d, if the number of experts of Astralaisa = y, then the number of experts of Americas = 2(y-1), the number of experts of Europe = y-1, the number of experts of Africa= y-1
Now, y+y-1+y-1+2(y-1)=21 => 5y-4=21 => y=5
Hence the table can be filled as follows:

	Africa	America	Australasia	Europe	Total
Labour					3
Health					6
Population Studies					6
Refugee Location					6
Total	4	8	5	4	21

Using condition b, the number of labour experts of Africa = 0 and the number of labour experts of the rest of the continents will be 1 each.
Consider the column of Europe, since Europe sent at least 1 in each category, each category will have 1 expert. (Since the total sum is 4)
From the condition e, the population studies of Australasia will contain 2 experts. So all the other categories will have to take at 1 expert because the total sum is 5 and no value can be 0. The table can be filled as follows:

	Africa	America	Australasia	Europe	Total
Labour	0	1	1	1	3
Health			1	1	6
Population Studies			2	1	6
Refugee Location			1	1	6
Total	4	8	5	4	21

Now, according to the condition c, the possible solutions are as given in the tables below:

	Africa	America	Australasia	Europe	(Total)
Labour	0	1	1	1	3
health	2	2	1	1	6
Population Studies	1	2	2	1	6
Refugee Location	1	3	1	1	6
(Total)	4	8	5	4	21

	Africa	America	Australasia	Europe	(Total)
Labour	0	1	1	1	3
health	1	3	1	1	6
Population Studies	1	2	2	1	6
Refugee Location	2	2	1	1	6
(Total)	4	8	5	4	21

	Africa	America	Australasia	Europe	(Total)
Labour	0	1	1	1	3
Health	1	3	1	1	6
Population Studies	2	1	2	1	6
Refugee Location	1	3	1	1	6
(Total)	4	8	5	4	21

From the above tables we can see that there was no possible case in which Africa and America had 1 expert each in population studies. Thus, statement D is false.

VIDEO SOLUTION

36. **C**

From the first condition a, assume the number of experts in labour = x, then the number of experts in each of the other three categories =2x

Now, x+2x+2x+2x+2x = 21 => x = 3 and 2x = 6

So the number of experts in labour = 3 and the number of experts in each of the three categories = 6

From condition d, if the number of experts of Astralaisa = y, then the number of experts of Americas = 2(y-1), the number of experts of Europe = y-1, the number of experts of Africa= y-1

Now, y+y-1+y-1+2(y-1)=21 => 5y-4=21 => y=5

Hence the table can be filled as follows:

	Africa	America	Australasia	Europe	Total
Labour					3
Health					6
Population Studies					6
Refugee Location					6
Total	4	8	5	4	21

Using condition b, the number of labour experts of Africa = 0 and the number of labour experts of the rest of the continents will be 1 each.

Consider the column of Europe, since Europe sent at least 1 in each category, each category will have 1 expert. (Since the total sum is 4)

From the condition e, the population studies of Australasia will contain 2 experts. So all the other categories will have to take at 1 expert because the total sum is 5 and no value can be 0. The table can be filled as follows:

	Africa	America	Australasia	Europe	Total
Labour	0	1	1	1	3
Health			1	1	6
Population Studies			2	1	6
Refugee Location			1	1	6
Total	4	8	5	4	21

Now, according to the condition c, the possible solutions are as given in the tables below:

	Africa	America	Australasia	Europe	(Total)
Labour	0	1	1	1	3
health	2	2	1	1	6
Population Studies	1	2	2	1	6
Refugee Location	1	3	1	1	6
(Total)	4	8	5	4	21

	Africa	America	Australasia	Europe	(Total)
Labour	0	1	1	1	3
health	1	3	1	1	6
Population Studies	1	2	2	1	6
Refugee Location	2	2	1	1	6
(Total)	4	8	5	4	21

	Africa	America	Australasia	Europe	(Total)
Labour	0	1	1	1	3
Health	1	3	1	1	6
Population Studies	2	1	2	1	6
Refugee Location	1	3	1	1	6
(Total)	4	8	5	4	21

If an American expert in refugee relocation, was the first keynote speaker in the conference. Then apart from him there is atleast 1 and atmost 2 american expert in refugee relocation. Hence option C.

VIDEO SOLUTION

37. **D**

From the first condition a, assume the number of experts in labour = x, then the number of experts in each of the other three categories =2x
Now, x+2x+2x+2x+2x = 21 => x = 3 and 2x = 6
So the number of experts in labour = 3 and the number of experts in each of the three categories = 6
From condition d, if the number of experts of Astralaisa = y, then the number of experts of Americas = 2(y-1), the number of experts of Europe = y-1, the number of experts of Africa= y-1
Now, y+y-1+y-1+2(y-1)=21 => 5y-4=21 => y=5
Hence the table can be filled as follows:

	Africa	America	Australasia	Europe	Total
Labour					3
Health					6
Population Studies					6
Refugee Location					6
Total	4	8	5	4	21

Using condition b, the number of labour experts of Africa = 0 and the number of labour experts of the rest of the continents will be 1 each.
Consider the column of Europe, since Europe sent at least 1 in each category, each category will have 1 expert. (Since the total sum is 4)
From the condition e, the population studies of Australasia will contain 2 experts. So all the other categories will have to take at 1 expert because the total sum is 5 and no value can be 0. The table can be filled as follows:

	Africa	America	Australasia	Europe	Total
Labour	0	1	1	1	3
Health			1	1	6
Population Studies			2	1	6
Refugee Location			1	1	6
Total	4	8	5	4	21

Now, according to the condition c, the possible solutions are as given in the tables below:

	Africa	America	Australasia	Europe	(Total)
Labour	0	1	1	1	3
health	2	2	1	1	6
Population Studies	1	2	2	1	6
Refugee Location	1	3	1	1	6
(Total)	4	8	5	4	21

	Africa	America	Australasia	Europe	(Total)
Labour	0	1	1	1	3
health	1	3	1	1	6
Population Studies	1	2	2	1	6
Refugee Location	2	2	1	1	6
(Total)	4	8	5	4	21

	Africa	America	Australasia	Europe	(Total)
Labour	0	1	1	1	3
Health	1	3	1	1	6
Population Studies	2	1	2	1	6
Refugee Location	1	3	1	1	6
(Total)	4	8	5	4	21

We can see that there are 2 possibilities for number of experts in refugee relocation from Africa.Hence it cannot be determined.

VIDEO SOLUTION

38. **C**

From statement 1 and 2,Each team speaks exactly four languages. English and Chinese are spoken by all three teams, Basque and French by exactly two teams and the other languages by exactly one team, multiple options are possible.

In the following tables: A, B, C can be any team among Team 1,Team 2,Team 3.

A	B	C
English	English	English
Chinese	Chinese	Chinese
Basque	Basque	Arabic
French	French	Dutch

A	B	C
English	English	English
Chinese	Chinese	Chinese
Basque	Basque	Arabic
Dutch	French	French

A	B	C
English	English	English
Chinese	Chinese	Chinese
Basque	Basque	Dutch
Arabic	French	French

From the data given in the question, the person who speaks Arabic also speaks French. Thus the only option possible is 'Table 2'.

A	B	C
English	English	English
Chinese	Chinese	Chinese
Basque	Basque	Arabic
Dutch	French	French

According to statement 4, "Paula and Sally are together in exactly two teams."

Sally knows Basque, thus, she will be in group A and B, with Paula.

According to statement 5, Robert(Arabic) is in Team 1 and Quentin(Dutch) is in Team 3.

Thus, Group C is Team 1 and Group A is Team 3.

Team 3		Team 2		Team 1	
English	Paula, Sally, Quentin	English	Paula, Sally	English	Paula
Chinese	Paula	Chinese	Paula, Terence	Chinese	Paula, Terence
Basque	Sally	Basque	Sally	Arabic	Terence, Robert
Dutch	Quentin	French	Terence	French	Robert

From the table, the correct option is C.

VIDEO SOLUTION

39. **C**

From statement 1 and 2,Each team speaks exactly four languages. English and Chinese are spoken by all three teams, Basque and French by exactly two teams and the other languages by exactly one team, multiple options are possible.

In the following tables: A, B, C can be any team among Team 1,Team 2,Team 3.

A	B	C
English	English	English
Chinese	Chinese	Chinese
Basque	Basque	Arabic
French	French	Dutch

A	B	C
English	English	English
Chinese	Chinese	Chinese
Basque	Basque	Arabic
Dutch	French	French

A	B	C
English	English	English
Chinese	Chinese	Chinese
Basque	Basque	Dutch
Arabic	French	French

From the data given in the question, the person who speaks Arabic also speaks French. Thus the only option possible is 'Table 2'.

A	B	C
English	English	English
Chinese	Chinese	Chinese
Basque	Basque	Arabic
Dutch	French	French

According to statement 4, "Paula and Sally are together in exactly two teams."

Sally knows Basque, thus, she will be in group A and B, with Paula.

According to statement 5, Robert(Arabic) is in Team 1 and Quentin(Dutch) is in Team 3.

Thus, Group C is Team 1 and Group A is Team 3.

Team 3		Team 2		Team 1	
English	Paula, Sally, Quentin	English	Paula, Sally	English	Paula
Chinese	Paula	Chinese	Paula, Terence	Chinese	Paula, Terence
Basque	Sally	Basque	Sally	Arabic	Terence, Robert
Dutch	Quentin	French	Terence	French	Robert

From the table, the correct option is C.

VIDEO SOLUTION

40. **C**

From statement 1 and 2,Each team speaks exactly four languages. English and Chinese are spoken by all three teams, Basque and French by exactly two teams and the other languages by exactly one team, multiple options are possible.

In the following tables: A, B, C can be any team among Team 1,Team 2,Team 3.

A	B	C
English	English	English
Chinese	Chinese	Chinese
Basque	Basque	Arabic
French	French	Dutch

A	B	C
English	English	English
Chinese	Chinese	Chinese
Basque	Basque	Arabic
Dutch	French	French

A	B	C
English	English	English
Chinese	Chinese	Chinese
Basque	Basque	Dutch
Arabic	French	French

From the data given in the question, the person who speaks Arabic also speaks French. Thus the only option possible is 'Table 2'.

A	B	C
English	English	English
Chinese	Chinese	Chinese
Basque	Basque	Arabic
Dutch	French	French

According to statement 4, "Paula and Sally are together in exactly two teams."

Sally knows Basque, thus, she will be in group A and B, with Paula.

According to statement 5, Robert(Arabic) is in Team 1 and Quentin(Dutch) is in Team 3.

Thus, Group C is Team 1 and Group A is Team 3.

Team 3		Team 2		Team 1	
English	Paula, Sally, Quentin	English	Paula, Sally	English	Paula
Chinese	Paula	Chinese	Paula, Terence	Chinese	Paula, Terence
Basque	Sally	Basque	Sally	Arabic	Terence, Robert
Dutch	Quentin	French	Terence	French	Robert

From the table, the correct option is C.

VIDEO SOLUTION

41. **A**

From statement 1 and 2,Each team speaks exactly four languages. English and Chinese are spoken by all three teams, Basque and French by exactly two teams and the other languages by exactly one team, multiple options are possible.

In the following tables: A, B, C can be any team among Team 1,Team 2,Team 3.

A	B	C
English	English	English
Chinese	Chinese	Chinese
Basque	Basque	Arabic
French	French	Dutch

A	B	C
English	English	English
Chinese	Chinese	Chinese
Basque	Basque	Arabic
Dutch	French	French

A	B	C
English	English	English
Chinese	Chinese	Chinese
Basque	Basque	Dutch
Arabic	French	French

From the data given in the question, the person who speaks Arabic also speaks French. Thus the only option possible is 'Table 2'.

A	B	C
English	English	English
Chinese	Chinese	Chinese
Basque	Basque	Arabic
Dutch	French	French

According to statement 4, "Paula and Sally are together in exactly two teams."

Sally knows Basque, thus, she will be in group A and B, with Paula.

According to statement 5, Robert(Arabic) is in Team 1 and Quentin(Dutch) is in Team 3.

Thus, Group C is Team 1 and Group A is Team 3.

Team 3		Team 2		Team 1	
English	Paula, Sally, Quentin	English	Paula, Sally	English	Paula
Chinese	Paula	Chinese	Paula, Terence	Chinese	Paula, Terence
Basque	Sally	Basque	Sally	Arabic	Terence, Robert
Dutch	Quentin	French	Terence	French	Robert

From the table, the correct option is A.

VIDEO SOLUTION